

Linux CLI Basics :- (Get Comfortable navigating through linux CLI)

In cybersecurity, linux is everywhere, powering servers, security tools and even hacking environments. However, before you can defend systems or investigate incidents, you need to know how to navigate linux OS using command line interface (CLI). This room takes on as brand new intern in a cyber security team. We will explore basic of linux CLI through hands on tasks, simple missions.

Learning Objectives :-

- * Understand what linux terminal is and what its used for
- * Feel comfortable interacting with linux terminal
- * Navigate linux file system with basic commands.

Storyline :- your first Day on Job :-

Congratulations, today is your first day as IT support Engineer with Cyber operations Support team.

Your Supervisor left in hurry deal with urgent incident. Before they lean out the door, they left you short note on your desk.

Most of work happens in linux terminal, and I've set up few simple tasks on system to help you get comfortable.

Quiz :- 1) what does "CLI" stand for? A) Command-line Interface

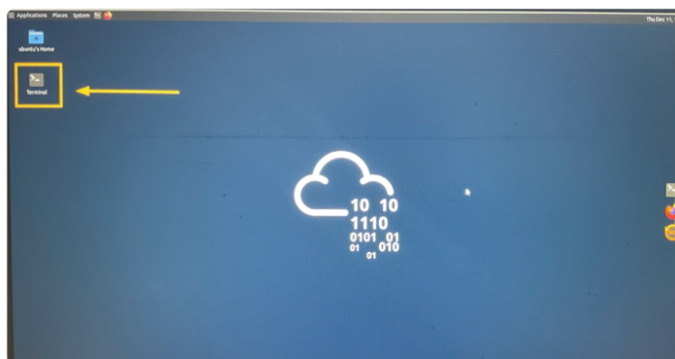
Task :- Navigation Mission : "Find the Missing Notes"

A Quick Note About Terminal :-

The Terminal is text based Interface for controlling a linux system. Instead of interacting with a graphical interface, you type commands that tell computer exactly what to do.

* Its faster than clicking around | It gives more control | * Many security tools only run in terminal.

By the end of room, you will feel far more comfortable. Double click Terminal file present on desktop.



Interaction with the Terminal :-

Your Supervisor mentioned leaving few tasks in document `mission-brief.txt`, Somewhere on System, means your first mission to find them. To do that, you'll need to learn how to move around Linux file system.

Step 1:- "Where AM I?"

Use this Command to find out: `Pwd`

Ubuntu@toyhack me: ~\$ `Pwd` } It stands for "present working directory" basically means
`/home/ubuntu` } "Show me the folder I am currently in".

Step 2:- "What's Around Me?"

Now that you know where you are, let's see what files and folders here: `ls`

Ubuntu@toyhack me: \$ `ls` } This lists the content of current directory. If we need
Desktop Downloads Pictures Templates logs } more details, we can try `ls -l`
Documents Music public videos projects

```
ubuntu@toyhackme:~$ ls -l
total 44
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Desktop
drwxr-xr-x 6 ubuntu ubuntu 4096 Dec 11 12:45 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 16 2024 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Videos
drwxr-xr-x 4 ubuntu ubuntu 4096 Dec 11 12:29 logs
drwxr-xr-x 5 ubuntu ubuntu 4096 Dec 11 12:29 projects
drwx----- 3 ubuntu ubuntu 4096 Sep 12 2024 snap
```

This output displays important information about files and directories like file sizes, permissions, dates and more.

Hidden files:- In order to get hidden files in directory, we can append command to `ls -al`, and it will display all hidden files present in directory, as shown below.

```
ubuntu@toyhackme:~$ ls -al
total 144
drwxr-xr-x 24 ubuntu ubuntu 4096 Feb 10 10:48 .
drwxr-xr-x 3 root root 4096 Feb 10 10:36 ..
-rw-r--r-- 1 ubuntu ubuntu 439 Feb 10 06:47 .Xauthority
-rw-r--r-- 1 ubuntu ubuntu 0 Sep 12 2024 .Xresources
-rw-r--r-- 1 ubuntu ubuntu 111 Oct 9 2024 .report-ignore.xml
.....
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Videos
drwxr-xr-x 4 ubuntu ubuntu 4096 Dec 11 12:29 logs
drwxr-xr-x 5 ubuntu ubuntu 4096 Dec 11 12:29 projects
drwx----- 3 ubuntu ubuntu 4096 Sep 12 2024 snap
```

Hidden files aren't really secret, they start with a dot, and Linux hides such files by default.

Step 3:- Let's Move Around

To walk through file system, we can use `cd <directory>`. For example: `cd Documents`, and this will change our directory to `Documents`, as shown around.

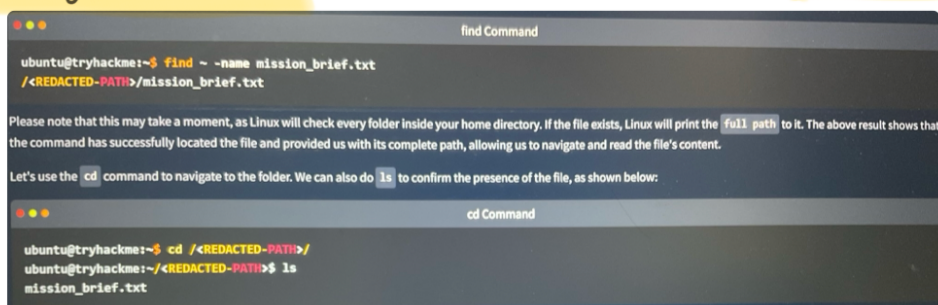
Ubuntu@toyhack me: ~\$ `cd Documents` } To go back one level, we will use command
ubuntu@toyhack me: ~/Documents\$ `Pwd` } `cd..` as shown below
`/home/ubuntu/Documents`

ubuntu@tryhackme: ~ /Documents\$ cd ..

ubuntu@tryhackme: ~\$ pwd | outcome :- /home/ubuntu

Step 4:- learn the power of "Find"

lets now use one of handy tools to Find. As name suggest built in utility is used to locate files within the file system. `find <Starting point> -name <filename>`. Since your supervisor mentioned the file `mission_brief.txt` resides somewhere in your home directory, begin the home directory symbol: `~`. so we will run the command: `find ~ -name mission_brief.txt`



```
ubuntu@tryhackme:~$ find ~ -name mission_brief.txt
/<REDACTED-PATH>/mission_brief.txt

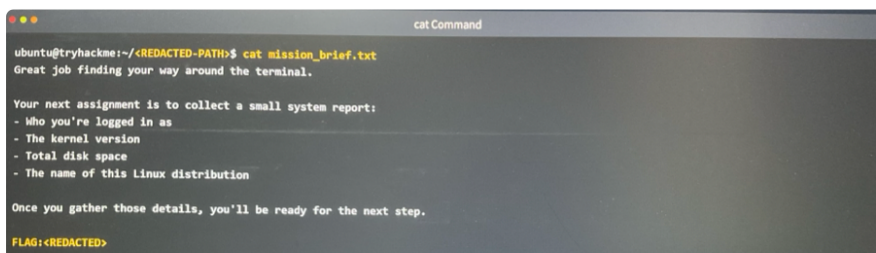
Please note that this may take a moment, as Linux will check every folder inside your home directory. If the file exists, Linux will print the full path to it. The above result shows that the command has successfully located the file and provided us with its complete path, allowing us to navigate and read the file's content.

Let's use the cd command to navigate to the folder. We can also do ls to confirm the presence of the file, as shown below:
```

```
ubuntu@tryhackme:~$ cd /<REDACTED-PATH>/
ubuntu@tryhackme:~/<REDACTED-PATH>$ ls
mission_brief.txt
```

Step 5:- Read the File

Another useful utility is `Cat`. This is used to read content of file. To read this file, we will run the command `cat mission_brief.txt` to get content, as shown below.



```
ubuntu@tryhackme:~/<REDACTED-PATH>$ cat mission_brief.txt
Great job finding your way around the terminal.

Your next assignment is to collect a small system report:
- Who you're logged in as
- The kernel version
- Total disk space
- The name of this Linux distribution

Once you gather those details, you'll be ready for the next step.

FLAG:<REDACTED>
```

lets proceed to next task and learn how to retrieve system information using commands in CLI.